MATH 463 - INTEGRAL EQUATIONS

LESSON 1

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OUTLINE

1. INTRODUCTORY CONCEPTS
   - Introduction to Integral Equations
   - Differentiation Under the Integral Sign
2. CLASSIFICATION OF INTEGRAL EQUATIONS
   - Structure
   - Kind
   - Linearity
   - Singularity
   - Homogeneity
3. SOLUTION OF AN INTEGRAL EQUATION

INTRODUCTION TO INTEGRAL EQUATIONS

DEFINITION: INTEGRAL EQUATION

An Integral Equation (IE) is an equation in which an unknown function appears under one or more integral signs.

This concept of integral equation was introduced by du Bois-Reymond in 1888. Before that however, there had been equations of this form. A classical example is the Laplace equation in 1782 which used the integral

$$f(x) = \int_0^\infty e^{-xu}\phi(u)\, du \tag{1}$$

GENERAL FORM

An integral equation is of the form

$$\mu(x) = f(x) + \lambda \int_{\alpha(x)}^{\beta(x)} K(x,t)\mu(t)\ dt \tag{2}$$

- $\mu(x)$ = unknown function: the solution we seek to determine
- $f(x)$ = free term / forcing term: it is known and continuous
- λ = investigative parameter: determines the variations of the problem/solution as λ is allowed to vary
- $\alpha(x), \beta(x)$ = limits of integration. They maybe constants or functions
- $K(x,t)$ = Kernel: it is known and continuous.

EXAMPLES

$$3x = \int_0^2 3xt\ y(t)\ dt$$

$$y(x) = 3x + \int_0^2 3xt\ y(t)\ dt$$

$$3x = \int_0^x \sin(3xt)\ y(t)\ dt$$

KERNEL

- The Kernel is a known and continuous function defined in the square $R : a \leq x \leq b, a \leq t \leq b$ of the $(x - t)$ plane
- A separable kernel can be expressed in the form $k(x, t) = g(x)h(t)$
- A degenerate kernel can be expressed in the form $k(x, t) = \sum_{i=0}^{n} g_i(x)h_i(t)$

- A symmetric kernel can be expressed in the form $k(x,t) = k(t,x)$
  1. A kernel of this property has at least one eigen value
  2. All the eigen values associated with the kernel are real.
  3. Eigen functions $\phi(x)$ and $\mu(x)$ of a symmetric kernel corresponding to distinct eigenvalues $\lambda 1$ and $\lambda 2$ ($\lambda 1 \neq \lambda 2$) are orthogonal

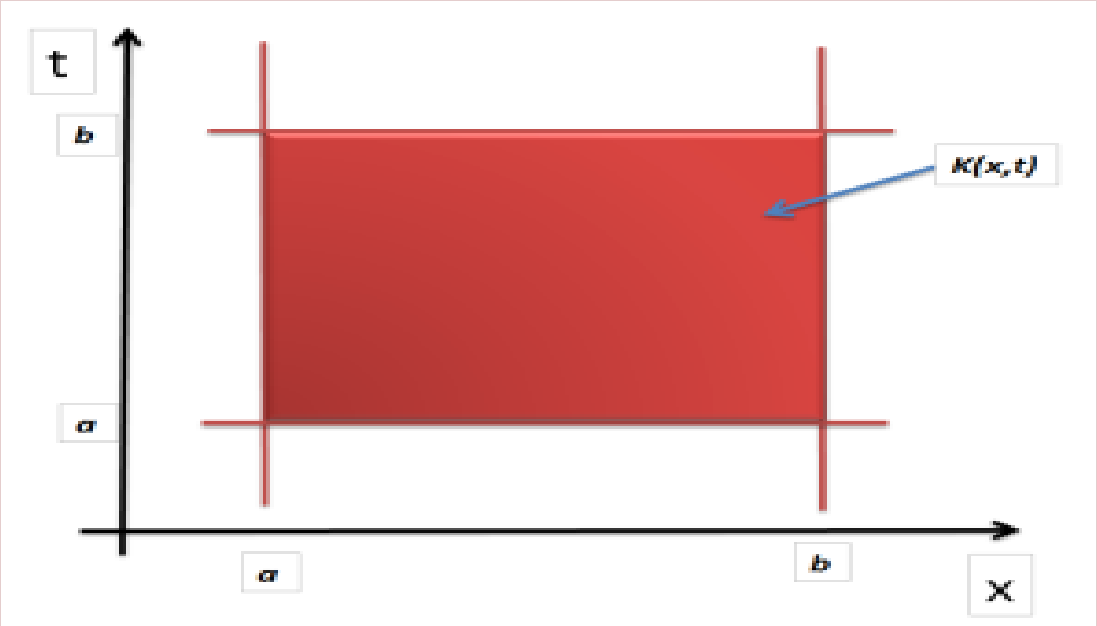

Figure: Kernel

DIFFERENTIATION UNDER THE INTEGRAL SIGN

In calculus, the rule for differentiation under the integral sign, called the Leibniz Integral Rule is named after Gottfried Wilhelm Leibniz, states that for an integral of the form

$$\int_{a(x)}^{b(x)} f(x,t)\ dt$$

where $-\infty < a(x), b(x) < \infty$ and the integrands are functions dependent on x, the derivative of this integral can be expressed by

Leibniz Integral Rule

$$\frac{d}{dx}\left(\int_{a(x)}^{b(x)} f(x,t)dt\right) = f(x,b(x)) \cdot \frac{d}{dx}b(x) - f(x,a(x)) \cdot \frac{d}{dx}a(x) \quad (3)$$

$$+ \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} f(x,t)\ dt \quad (4)$$

Example using Leibniz Integral Rule

Differentiate the following integral

$$\frac{d}{dx}\int_0^x 3xt\ dt$$

Solution

$$\frac{d}{dx}\int_0^x 3xt\ dt$$

$$= 3x(x)\frac{dx}{dx} - 3x(0)\frac{d0}{dx} + \int_0^x \frac{\partial(3xt)}{\partial}\ dt$$

$$= 3x^2 - 0 + \int_0^x 3t\ dt = \frac{9x^2}{2}$$

Exercises

1. State the free term, investigative parameter and the kernel of the following integral equations.
   1. $$u(x) = \cos(x) + \int_0^2 \cos(x)u(t)\ dt$$
   2. $$u(x) = \int_0^x t^2 u^2(t)\ dt$$
2. After identifying the kernels in the integral equations above, state whether there are separable or degenerate.
3. Solve
$$\frac{d}{dx}\int_0^x k(x,t)u(t)\ dt$$

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Introduction

There are various ways of classifying integral equations. We can classify them by their structure: whether Fredholm or Volterra; by their kind: whether 1st kind or 2nd kind; by linearity: whether linear or non linear; by singularity: whether singular or non singular; by homogeneity: whether homogeneous or non homogeneous.

STRUCTURE

VOLTERRA INTEGRAL EQUATION

The limits of integration contains a function of x.
example:

$$\mu(x) = \cos(x) + 2\int_0^x \sin(xt)\mu(t)\ dt$$

FREDHOLM'S INTEGRAL EQUATION

The limits of integration contains constants only.
example:

$$\mu(x) = \cos(x) + 2\int_0^2 \sin(xt)\mu(t)\ dt$$

KIND

1ST KIND

If the unknown function appears only under the integral sign.
example:

$$\cos(x) = 2\int_0^x \sin(xt)\mu(t)\ dt$$

2ND KIND

If the unknown function appears under the integral sign and also at L.H.S.
example:

$$\mu(x) = \cos(x) + 2\int_0^2 \sin(xt)\mu(t)\ dt$$

LINEARITY

It depends on the nature of the unknown function

LINEAR

If the unknown function does not exist as a trancendental function and has a power of 1.
example:

$$\cos(x) = 2\int_0^x \sin(xt)\mu(t)\ dt$$

NON-LINEAR

If the unknown function exist as a trancendental function and or has a power of 2 or more.
example:

$$\mu(x) = \cos(x) + 2\int_0^2 \sin(xt)\mu^2(t)\ dt$$

SINGULAR

SINGULAR

When the integral involved is an improper integral
example:

$$\cos(x) = 2\int_0^{\infty} \sin(xt)\mu(t)\ dt$$

NON-SINGULAR

example:

$$\mu(x) = \cos(x) + 2\int_0^{2} \sin(xt)\mu(t)\ dt$$

HOMOGENEITY

HOMOGENEOUS

When the free term is equivalent to 0

example:

$$\mu(x) = 2\int_0^2 \sin(xt)\mu(t)\ dt$$

NON-HOMOGENEOUS

example:

$$\mu(x) = \cos(x) + 2\int_0^2 \sin(xt)\mu(t)\ dt$$

Exercises

1. State the structure, kind, linearity, singularity, and homogeneity of the following integral equations.
   1. $$u(x) = \cos(x) + \int_0^2 \cos(x)u(t)\ dt$$
   2. $$u(x) = \int_0^x t^2 u^2(t)\ dt$$

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SOLUTION OF AN INTEGRAL EQUATION

A solution of an integral equation is a function which when substituted into the equation, registers an identity.

- Show that $y(x) = x$ is a solution to the integral equation,

$$y(x) = \frac{5}{6}x - \frac{1}{9} + \frac{1}{3}\int_0^1 (x+t)y(t)\ dt$$

$$y(x) = x, y(t) = t$$

$$x = \frac{5}{6}x - \frac{1}{9} + \frac{1}{3}\int_0^1 (x+t)t\ dt$$

$$x = \frac{5}{6}x - \frac{1}{9} + \frac{1}{3}\int_0^1 xt + t^2\ dt$$

$$x = \frac{5}{6}x - \frac{1}{9} + \frac{x}{6} + \frac{1}{9}$$

$$x = x$$

Exercises

1. State the structure, kind, linearity, singularity, and homogeneity of the following integral equations and verify if the following given solutions are actual solutions of the respective integral equations.
   1. $$u(x) = x + \int_0^{\frac{1}{4}} u(t)\ dt; \qquad u(x) = x + 124$$
   2. $$u(x) = 2\cosh(x) - x\sinh(x) - 1 + \int_0^x tu(t)\ dt; \qquad u(x) = \cosh(x)$$
2. If $u(x) = x^2 + x^3$ is a solution to the integral equation below, then find λ
$$u(x) = x^3 - x^2 - 2x + \lambda \int_{-1}^{1} (xt^2 + tx^2)u(t)\ dt$$